CS-137 Assignment 2: Convolutional Neural Networks for Weather Forecasting

Goal

The goal of this assignment is to gain hands-on experience with **Convolutional Neural Networks (CNNs)** and to practice solving near-real-world problems using deep learning. You will work with real meteorological data to build a model that forecasts local weather conditions.

Project Description

In this project, you will build a CNN-based model that takes a snapshot of the weather at time **t** and predicts weather conditions 24 hours later (**t + 24h**) at the grid point nearest to the location of the **Jumbo Statue at Tufts University** (42.40777867717294, -71.12041637590173).

Input

The input **X_t** is a spatial snapshot of weather variables at time t, with shape:

`(450, 449, c)`

where `c` is the number of weather variables over the spatial area. The spatia average covers New England area of the US, which is shown by the figure below.

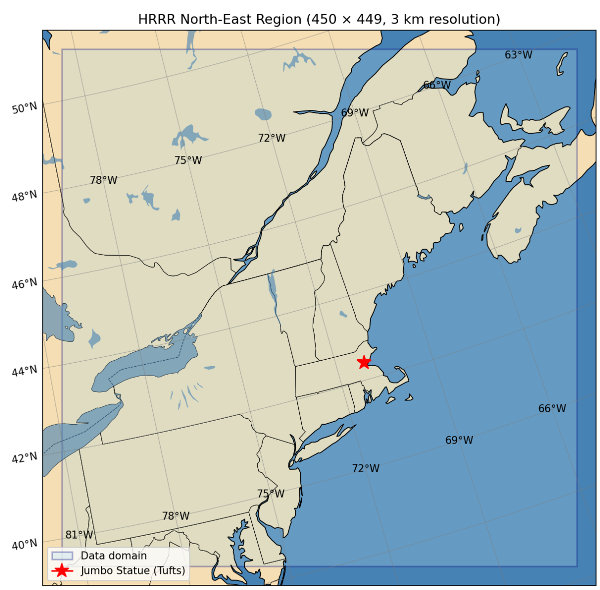

Figure 1: Spatial area of the study

Prediction Targets

The model should predict the following six weather variables at the target grid point 24 hours ahead:

Variable	Unit
`TMP@2m_above_ground`	K
`RH@2m_above_ground`	%
`UGRD@10m_above_ground`	m/s
`VGRD@10m_above_ground`	m/s
`GUST@surface`	m/s
`APCP_1hr_acc_fcst@surface`	mm
`APCP_1hr_acc_fcst@surface > 2mm`	binary label

Evaluation

- All continuous target variables are evaluated using **Root Mean Squared Error (RMSE)**.
- The RMSE for `APCP_1hr_acc_fcst@surface` is computed **only when the true value exceeds 2 mm**.
- The binary label (`APCP_1hr_acc_fcst@surface > 2mm`) is evaluated using the AUC (area under the ROC curve).

Data Preparation

All the data are at `/cluster/tufts/c26sp1cs0137/data/assignment2_data`. Currently the dataset covers **three years** of hourly weather data. We will add new data to the dataset. Two data files are provided in the `dataset/` folder:

1. **Input tensors** — For each hour *t*, all weather variables over the spatial area are stored as a `torch.Tensor` of shape `(450, 449, c)` with dtype 'bfloat16. Some input tensors contain NaN values. Please just neglect these tensors.
2. **Target tensors** — Target values at the Jumbo Statue grid point for all timesteps, including the binary precipitation label.

A Jupyter notebook is also provided with example visualizations, including: - A map of the spatial region - A plot of temperature over the spatial area - A plot of average temperature over one year

Grading (100 Points)

Part 1 — Train a CNN (60 points)

Successfully train a CNN model that takes the spatial weather snapshot as input and predicts the six target variables 24 hours ahead. You are free to choose your architecture, but your model must:

- Accept inputs of shape `(450, 449, c)`
- Predict all six target variables for the Jumbo Statue location
- Be evaluated on the held-out test set using RMSE (and classification metric for the binary label)
- Include a brief write-up describing your architecture and training procedure

Part 2 — Diagnosing Where the Weather Comes From (20 points)

Study and explain **which regions of the input map most influence the model's predictions**. Specifically:

- Compute and visualize the **gradient of the output with respect to the input** (saliency maps / sensitivity analysis)
- Discuss what the gradients reveal about which geographic areas or weather patterns drive the forecast
- Relate your findings to physical intuition about weather systems (e.g., prevailing wind directions, upstream regions)

Part 3 — Independent Study at Choice (20 points)

Choose **one** of the following topics to investigate and report on:

Option	Description
Architecture choice	Compare different CNN architectures (e.g., depth, kernel sizes, pooling strategies) and analyze their effect on forecast accuracy
Convolutional kernel diagnosis	Visualize and interpret learned convolutional filters; discuss what spatial patterns they detect
Input variable study	Perform ablation experiments to determine which input variables contribute most to predictive skill
Residual connections (ResLinks)	Study the effect of adding residual (skip) connections on model performance and training dynamics
Pre-training from generic data	Investigate whether pre-training the CNN on a broader dataset (e.g., a different region or time period) before fine-tuning improves results

Your write-up for Part 3 should include clear experiments, results, and discussion.

Working in Groups

Students **must** work in groups. The maximum group size is **3 students**. Each group submits a single report and codebase. All group members are expected to contribute and should briefly describe their individual contributions in the submission.

Deliverables

- **Code**: A well-organized repository or zip file containing all training scripts and notebooks
- **Report**: A written report (PDF or Markdown) covering Parts 1, 2, and 3
- **Model checkpoint**: Your best trained model weights

Good luck, and have fun forecasting the weather at Jumbo's location!